LESSON 9 • TERM 1 • LEVEL 1

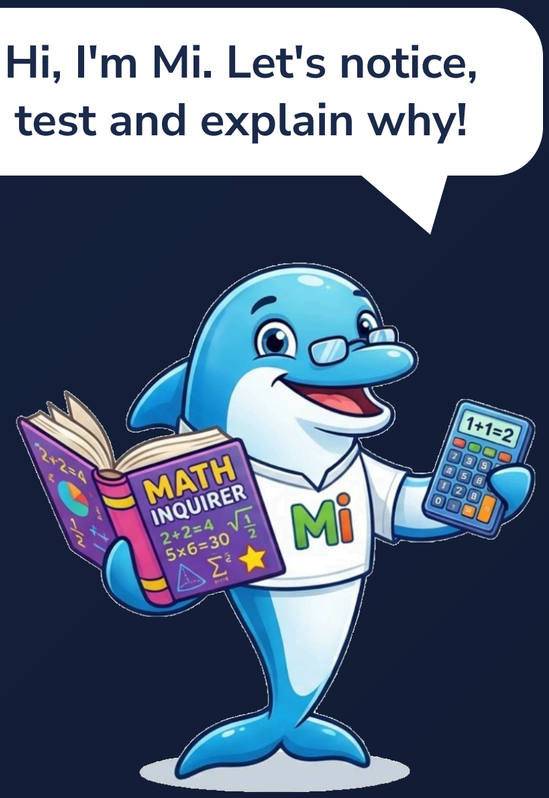
Draw and Measure

Length: centimetres

TODAY WE WILL

Use a ruler to measure and to draw lines exactly.





4–10 MIN

Warm-up: measure with clips

1. Lay clips along your pencil, end to end.
2. No gaps. No overlaps.
3. Count the clips.

WHAT IF?

Your friend uses bigger clips. Will the number be the same?

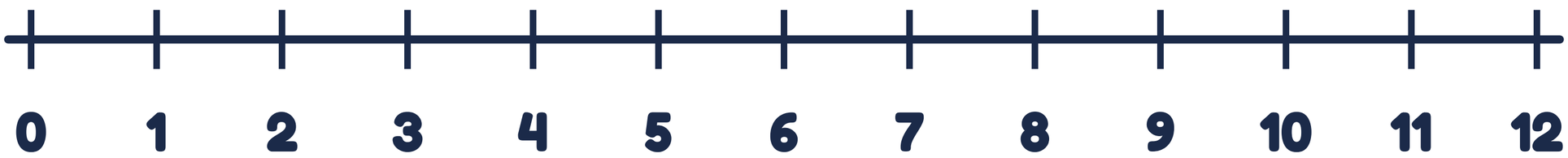


Rulers ready? Off we go!

4–10 MIN

Find 0 on your ruler

Put your finger on the 0 mark. Count along the marks.

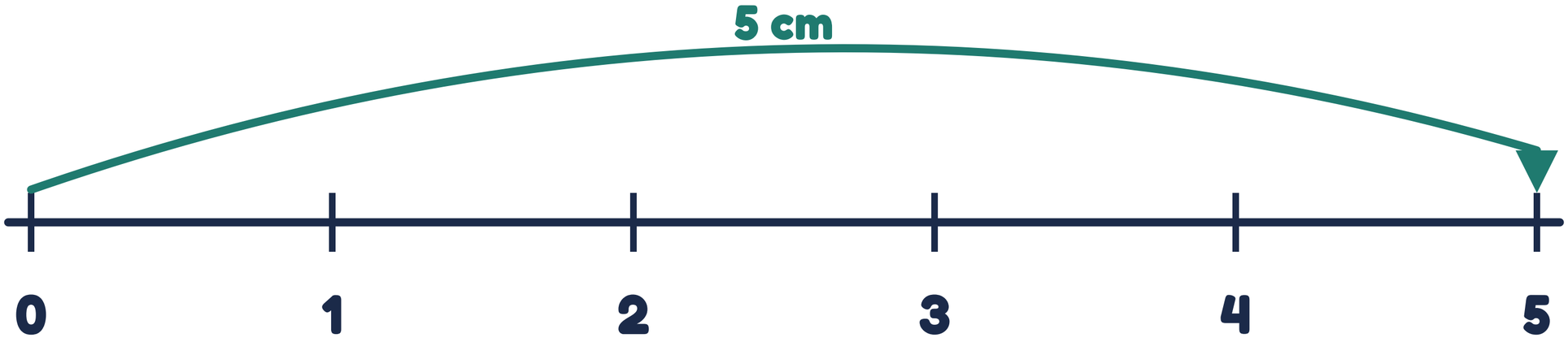


Say the numbers with me

4–10 MIN

One centimetre

The gap between two marks is 1 cm.



Slide your finger from 0 to 5. How many gaps? How many cm?

Sneak-peek puzzles

10–14 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

Draw a dot.

Use your ruler.

Draw a line from the dot, exactly 3 cm long.

PUZZLE 2: think

Ali puts one end of a pencil at the 1 mark of his ruler.

The other end is at the 9 mark.

Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.

Explore: guess, then measure

14–28 MIN

1. Find 3 things: a pencil, a spoon, a rubber or a toy.
2. Guess how many cm long each one is.
3. Put one end at the 0 mark. Read the mark at the other end.
4. Say: my ___ is about ___ cm.

WHAT IF?

The end is between two marks. Which mark is closer?

14–28 MIN

Draw it!

1. Put a dot where the line starts.
2. Put the 0 mark on the dot.
3. Make a small mark at the end.
4. Join them with a straight line.

3 cm | **8 cm** | **12 cm**

14–28 MIN

What did we notice?

- Where do we start measuring?
- What do we do when the end is between two marks?
- Why do we all use centimetres?

The parts of a ruler

28–38 MIN



REMEMBER

0 is where we start.

The gap from one mark to the next is 1 cm.

28–38 MIN

Measure: three steps

1. Put one end of the thing at the 0 mark.
2. Keep the ruler straight along the thing.
3. Read the mark at the other end. That is the length in cm.

Look from straight above

28–38 MIN

Draw a line 7 cm long

1. Put a dot where the line starts.
2. Put the 0 mark on the dot.
3. Make a small mark at the 7.
4. Join them with a straight line.
5. Check: 0 at the start, 7 at the end.

28–38 MIN

Between two marks

The end is between the 8 mark and the 9 mark. It is closer to the 9 mark.

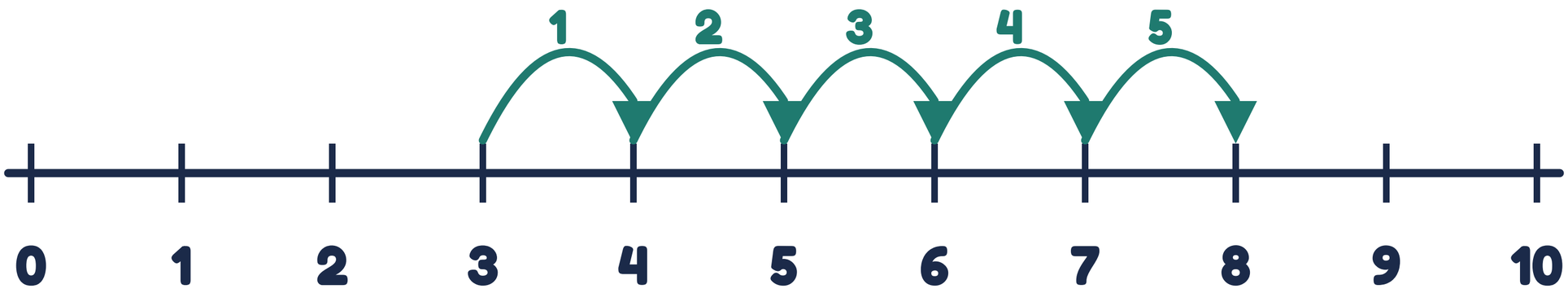


NEAREST CENTIMETRE

Choose the closer mark. This one is about 9 cm.

28–38 MIN

A line that starts at 3



The line starts at 3 and ends at 8. Count the jumps. 5 jumps. The line is 5 cm long.

8 take away 3 is 5

28–38 MIN

Your turn

Write on your paper. About 1 minute

DRAW

Draw a line 5 cm long.

WORK OUT

A line goes from the 2 mark to the 9 mark.

How long is it?

28–38 MIN

Check: your turn

DRAW

The line starts at the 0 mark and ends at the 5 mark.

WORK OUT

9 take away 2 is 7.

7 cm

38–42 MIN

Jump the tens, hop the ones

Pretend a big jump is 10 cm. A small hop is 1 cm.

14 23 30 8

14 cm is 1 big jump and 4 small hops. Sip some water.

42–70 MIN

Workbook page 1: Start here

Questions 1 to 3. About 9 minutes. Use your ruler

S1 In the box, draw a line 7 cm long. Start at the 0 mark.

S2 Draw a line 4 cm long. Under it, draw a line 9 cm long.

S3 Measure the length of your pencil. Write your guess first. Then measure to the nearest cm.

Check page 1

42–70 MIN

S1	A line from the 0 mark to the 7 mark: it starts at 0, ends at the 7 mark and is 7 cm long
S2	9 cm; it is 5 cm longer
S3	OPEN ANSWER, accept any sensible measure (guess and measure within about 3 cm of each other); a new pencil is about 17 to 19 cm. Check: the 0 mark at one end, read the mark at the other end

Workbook page 2: Start here

42–70 MIN

Questions 4 to 7. About 9 minutes. Use your ruler

S4 Ali's pencil and Mei's pencil both start at the 0 mark of a ruler. Ali's tip is at the 13 mark. Mei's tip is at the 9 mark.

S5 Draw a line 5 cm long. Under it, draw a line 3 cm longer than the first line.

S6 Mei's line goes from the 0 mark to the 7 mark. Mei says it is 8 cm long.

S7 The tip of Ben's spoon is between the 14 mark and the 15 mark. It is closer to the 15 mark. The spoon starts at the 0 mark.

Check page 2

42–70 MIN

S4	13 cm; 9 cm; Ali's is 4 cm longer
S5	The second line is 8 cm long (5 + 3)
S6	No. Her line is 7 cm. She should draw on to the 8 mark
S7	15 cm (the tip is closer to the 15 mark)

42–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Our idea in a new setting
Go further (page 4)	Bigger puzzles
Investigation (page 5)	Does the start mark matter?
Check	Finish and check your own work

Everyone may choose. You can swap at any time.

Talk and explain

70–76 MIN

MEI AND BEN MEASURE THE SAME PENCIL

Mei says 15 cm.

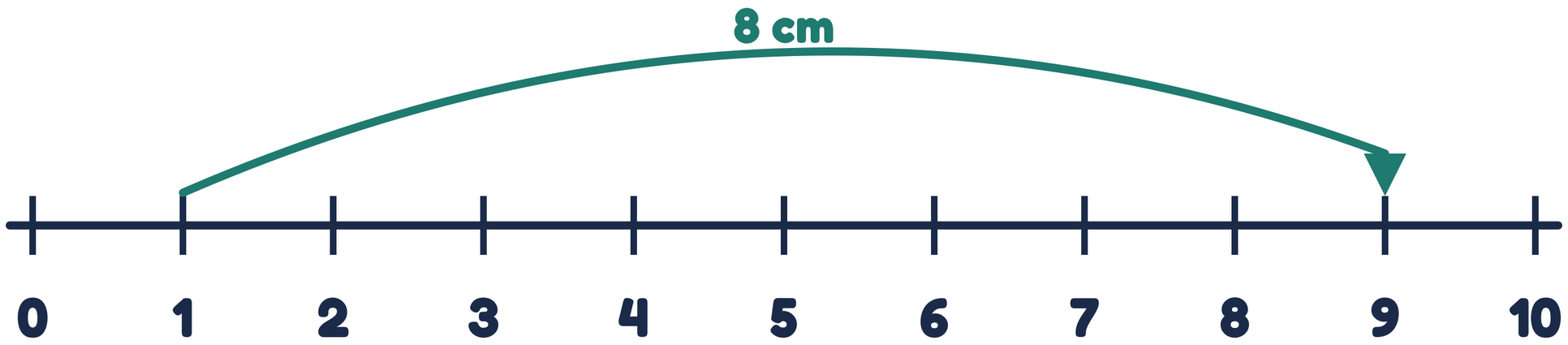
Ben says 16 cm.

Why might their answers be different? How can we check?

I think … because …

70–76 MIN

Back to the sneak-peek



No! The pencil starts at the 1 mark, so it is 8 cm long: 9 take away 1.

Exit ticket

76–82 MIN

On your own. No looking back. About 3 minutes

E1 Draw a line 6 cm long on your spare paper. Start it at the 0 mark of your ruler. Then fill in the marks.

E2 Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.

E3 A line starts at the 3 mark and ends at the 10 mark. How long is the line?

Check the exit ticket

76–82 MIN

E1	A line from the 0 mark to the 6 mark
E2	9 cm longer
E3	7 cm. 10 take away 3 is 7 (7 jumps from 3 to 10)

Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ starting at the 0 mark ☐ drawing a line exactly ☐ a line that starts at another mark ☐ adding the pieces of a path ☐ something else
What helped me	☐ my ruler ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck
Next time I will try	☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question
	☐ I am still stuck (my tutor will come and help) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Story Steps: Read It Twice

You need a pencil and paper.



OPTIONAL

Go further: how many cuts?

A strip of paper is 10 cm long. You may cut it at the 1 mark, the 2 mark and so on, but not at the ends.

How many places can you cut? What if the strip is 20 cm long?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	A line ends at the 9 mark. Is it always 9 cm long?
Make a puzzle	Draw a path with 3 straight pieces for your grown-up to measure
Teach it back	Show how to draw a line 9 cm long, in steps

Optional. Everyone may join.